Team Tarot Club
Mohameden Sidi-ahmed · Fuad Ahamed Jahubar Arif · Reema A A Awad · Nazer Alzaabi
Al Ain University · Cybersecurity, Computer Science · Abu Dhabi

STAND 27 · PROJECT ID 5007
EDGE TRACK · ATP 2026

FLEXIPATH

Switch before the link fails.

Someone is waiting for this link.
The drone must not go silent.



PROBLEM STATEMENT

“How might we keep a user's experience unbroken as they move between Wi-Fi, mobile, satellite and wired networks?”

EDGE Group challenge statement · C2 = command and control link · LEO = low Earth orbit

1 DETECT Watch signal, loss, delay and throughput.

2 PREDICT Find the time to failure. The map knows dead zones.

3 PREPARE Wake a spare radio. Test it with small packets.

4 REPLICATE Copy C2 on two paths for a short time.

5 SWITCH The tested path becomes primary.

6 CONTINUE Old radio sleeps. One address stays.

01 PROPOSED SOLUTION

A drone beyond its pilot's sight needs a control link (C2) that never stops. A **break-before-make** handover waits for the old network to fail. Then C2 is silent while the drone reconnects. In our model, a long silence makes the drone stop and hover.

FlexiPath is software under the drone's applications. When the active link gets weak, it wakes a spare radio and tests it. Then it moves C2 **before** the old link fails. A relay on the ground keeps one fixed address, so no session restarts.

Each part is known (MPTCP, 3GPP ATSSS, PDCP duplication). The combination for a drone on a planned route is our work.

02 TECHNICAL EXPLANATION

The engine measures signal, loss, delay and throughput on the active link. It fits a short trend to the signal and estimates when loss will become too high. A map from earlier flights warns of known dead zones. It tests the spare modem with small heartbeat messages. During the switch it copies C2 on both links; voice joins only when its loss is high. The receiver keeps the copy that arrives early and drops the other. All traffic goes through a UDP tunnel to an anchor relay.

A deterministic simulator and a 3D digital twin compare four stacks on the same seeded flights: legacy, reactive multipath, always duplicate, and FlexiPath. Detection and modem wake times are assumed values, not measurements.

03 SOLUTION FEATURES

- **Acts early:** it moves before a predicted failure.
- **Tested spare:** a sleeping modem is woken and tested before use.
- **C2 priority:** voice is copied only when needed. Video is never copied.
- **One address:** no session restarts.
- **Route foresight:** a coverage map warns of dead zones.
- **Power-aware:** spare radios sleep until needed.
- **Explainable:** a straight-line trend, not machine learning. Each decision shows its reason.
- **Known costs:** it can move early to a slower satellite link. The copy window shows no measured gain; the trend does the work.

04 FUTURE IMPROVEMENTS

- **Pilot:** Wi-Fi, 5G and LEO modems on a Linux companion computer, with one drone operator.
- **Need:** flight logs, a 5G test network, one LEO terminal.
- **Security:** an encrypted, authenticated tunnel and an approved-network list.
- **Better prediction:** tune the trend on real flight logs.
- **Fair tests:** equal detection times, Wi-Fi roaming, slow satellite links.

Built by four students in Al Ain.
We checked our own claims. We fixed what was wrong.

BUILT	SIMULATED	MISSING
the engine, a 3D twin, a benchmark	all radios, networks and results	real radios, encryption, flight tests

AI use: Claude (Anthropic) helped with code, documents, the video and this poster. Blender MCP built the 3D scene. The video voice is AI (Kokoro). The concept, design and result checks are ours. AI once called our hover limit a 3GPP rule; our review caught it and we fixed it.

SEE IT LIVE ON THE SCREENS BELOW

STATUS built, simulated, missing

LIVE RESULTS every number, with its conditions

LIVE DEMO the 3D twin